Chapter 4 - Functions and Continuity

For around 200 years, the term function simply meant analytic expression. That is a formula consisting of sums/products of polynomials, sines, cosines, square roots etc. By their very nature meant they were implitly smooth and didn't permit 'jumps' or gaps. This notion of function began to break down under the work of Fourier. The more modern definition of function we owe to Dirichlet.

Functions

Defn: A Function $f: S \to T$, with S and T sets, associates to each element $x \in S$ a unique element $f(x) \in T$. The set S is called the domain of f and the set T the co-domain of f.

Ex: By the above definition $f(x) = \pm\sqrt{x}$ is not a function.

Defn : Real function

A **real function** is a function with $\mathbb{R}$ as its co-domain.

Defn : i) Let $f: \mathbb{R}-\{0\} \to \mathbb{R}$ and $g: (1,\infty) \to \mathbb{R}$ given by

$$f(x) = \sin\left(\frac{1}{x}\right), \; g(x) = \sin\left(\frac{1}{x}\right).$$

Note that $f(x) \neq g(x)$ since the domains are different.

(ii) Functions can be defined **piecewise**. For example, the function $h: \mathbb{R} \to \mathbb{R}$ given by

$$h(x) = \begin{cases} \sin(x) : x \geq 0 \\ \cos(x) : x < 0 \end{cases}.$$

(iii) Let $d: \mathbb{R} \to \mathbb{R}$ given by

$$d(x) = \begin{cases} 1 : x \in \mathbb{Q} \\ 0 : x \notin \mathbb{Q} \end{cases}.$$

This function is called the **Dirichlet function**.

Defn: (Real Intervals)

An interval I is a subset of $\mathbb{R}$. They are either of th form

i) $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$ or

$(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$. These are clopen or half-open intervals.

ii) $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$ or

$[a, +\infty) = \{x \in \mathbb{R} : a \leq x\}$

iii) $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$, Closed

iv) $(a, b) = \{x \in \mathbb{R} : a < x < b\}$

$(-\infty, +\infty) = \mathbb{R}$

$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$ Open

$(a, +\infty) = \{x \in \mathbb{R} : a < x\}$.

In each case, it was assumed $a < b$ with $a, b \in \mathbb{R}$.

New functions from Old

Defn: Algebra on Real Functions

Let $f: S_1 \to \mathbb{R}$ and $g: S_2 \to \mathbb{R}$. Define $S := S_1 \cap S_2$. Then, the functions $f+g$, fg, f/g and cf are defined pointwise by:

(i) $(f+g)(x) = f(x) + g(x) \quad \forall x \in S$

(ii) $(fg)(x) = f(x)\,g(x) \quad \forall x \in S$

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ providing $g(x) \neq 0 \quad \forall x \in S$

(iv) $(cf)(x) = c f(x) \quad \forall c \in \mathbb{R}$ and $x \in S$.

Additionally, the function $|f| : S_1 \to \mathbb{R}$ is defined by

$$|f|(x) = |f(x)|$$

$\forall x \in S_1$.

Defn: Composition of Functions

Let $f: S \to D$ and $g: D \to T$. Then the composition $h: S \to T$ of g and f is given by

$$h(x) = g(f(x))$$

and is denoted by $h = g \circ f$.

Defn: Bounded Functions

A real function $f: S \to \mathbb{R}$ is:

(i) bounded above if $\exists K_u \in \mathbb{R}$ such that

$$f(x) \leq K_u$$

$\forall x \in S$. The number K_u an upper bound for f.

(ii) bounded below if $\exists K_L \in \mathbb{R}$ s.t

$$f(x) \geq K_L \quad \forall x \in S.$$

The number K_L is a lower bound for f.

(iii) bounded if it is bounded above and below.

If f is not bounded, we say its unbounded.